

Mapping Community Values in the Social Impacts of Computing (SioC 2025 Workshop)

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Contents

1	Introduction	2
2	Methods	3
3	Network Analysis	7
4	Discussion	10
5	Conclusion	13

1. Introduction

Emerging technologies like artificial intelligence (AI), extended reality (XR), and data-intensive systems are playing an increasingly influential role in human society. These technologies are reshaping how we interact and form relationships, how we learn and educate, how cultural behaviors spread, and even how we impact the environment. For example, the rapid integration of generative AI is already reshaping education – offering new opportunities to enhance learning and equity of access, while also introducing risks that demand new literacies and awareness (Gunder et al., 2025).

Likewise, XR is transforming immersive learning and collaboration, data-driven algorithms are influencing social behaviors, and high-performance computing is raising concerns about energy and environmental costs. Recognizing both the promise and perils of such technologies, there is growing interest in ensuring they contribute to human flourishing rather than impede it. The SioC 2025: Emerging Technologies and Well-Being workshop was convened around this very challenge – asking not only what these technologies can do, but how their design and social impacts (intended and unintended) might make our lives as workers, friends, learners, and citizens better. A key feature of the SioC 2025 workshop was its thematic focus on distinct domains of impact: XR and Educational Technology, Data Ethics, Environmental Impacts, and Healthcare. Across these domains, discussions implicitly invoke various social values – from equity and inclusion in ed-tech, to transparency and responsibility in data practices, to sustainability in environmental tech, to empathy and care in health innovations. Scholars and practitioners often assume certain values when debating ethical technology design (for instance, calling for transparency in AI or equity in access to XR tools). However, these assumptions are rarely grounded in empirical data about what community members themselves associate with the phrase “social impacts of computing.” This represents a research gap: we lack clear evidence of which social values and concerns are top-of-mind for stakeholders when they think about computing’s role in society. Prior work in cultural studies and technology literacy suggests that meanings and values are socially situated. Gunder et al. (2025), for example, emphasize aligning AI innovations with the needs and values of the open community, advocating an “opened culture” of technology use that fosters transparency, equity, and ethical governance (Gunder et al., 2025).

Similarly, Richerson et al. (2016) argue that human communities can develop distinct norms and values through cultural processes – following Darwin’s classic syllogism that if groups vary in their cultural traits (and those traits affect group success), then those traits (e.g. cooperative values) will spread in that group. In other words, shared values emerge through discourse, culture, and practice, not just top-down principles (Gunder et al., 2025). With this in mind, our study asks: What values do participants in the SioC community actually invoke when they talk about “the social impacts of computing”? By empirically mapping these values, we aim to advance the conversation beyond assumed ideals, toward a grounded understanding of what matters to this community in real-world discussions of technology and society.

2. Methods

To investigate community perceptions, we administered a pre-workshop survey to all SioC 2025 participants. This survey posed two main questions about the term “social impacts of computing”:

1. An open-ended prompt: “What does the term ‘social impacts of computing’ mean to you?” – where participants could write a free-text response in their own words.
2. A values checklist: “What values matter to you most when considering computing’s social impact?” – where participants selected from a list of given values (with an option to write in “Other” values not listed).

We received 82 responses to these questions. The data were analyzed using qualitative content analysis and thematic coding techniques. For the open-ended definitions (Question 1), we performed iterative open coding to identify recurring themes and concepts in how people described “social impacts of computing.” In parallel, we examined the choices in the values checklist (Question 2), which included predefined terms (e.g. Inclusivity, Transparency, Equity, etc.) as well as additional values that participants wrote in. We then synthesized these inputs to develop a refined set of ≈ 14 distinct values that captured the range of ideas expressed. This list was informed by our open coding, the checklist options, and an existing values framework (the AIROE - VAMPIRE codebook) to ensure we included both expected and emergent values. Example values in this final set included Responsibility, Transparency, Integrity, Inclusivity, Equity, Adaptability, Environmental Stewardship, Efficiency, Caring, Respect, Curiosity, and Growth, among others. Each participant’s response was coded to one or more of these value categories (for instance, a response might be tagged with Responsibility and Transparency if it mentioned ethical accountability and openness in computing).

Table 2.1: Codebook for SioC Values

Value	Definition
Adaptability	Flexibility and openness to change in the face of new computing developments. This value signifies willingness to adjust strategies and learn new skills as technology and social conditions evolve. An adaptable approach ensures that individuals and communities can handle challenges and opportunities brought by rapid technological change.
Caring	Demonstrating empathy and concern for people and communities affected by computing. Caring means using technology in ways that prioritize well-being and showing compassion – for example, designing inclusive tools and considering the human impact of computing decisions on others’ lives.

Table 2.1: Codebook for SioC Values (*continued*)

Value	Definition
Citizenship	Contributing positively to society and the community through computing. This value entails acting as a good “digital citizen” by considering the public good, promoting societal well-being and minimizing harm, and engaging in the civic implications of technology (e.g. addressing social needs or inequalities).
Collaboration	Valuing teamwork and cooperative problem-solving in computing contexts. This means working together with others openly and effectively, sharing knowledge, and combining skills to achieve common goals in technology projects or societal initiatives.
Curiosity	A value characterized by a desire to learn, explore, and understand new aspects of computing. Curiosity means being inquisitive about how technology works and its impacts, driving continuous learning and innovation. It reflects an intrinsic motivation to ask questions and seek knowledge in the computing domain.
Efficiency	Striving for optimal performance and resource use in computing. This means achieving goals with minimal waste of time or resources – for instance, automating tasks to save time. In computing’s social context, efficiency is often about saving effort and cost for individuals, organizations, or society through smart use of technology.
Environmental Stewardship	Commitment to minimize the ecological footprint of computing and protect the environment. This value involves considering energy consumption, e-waste, and climate impacts of technology. It reflects environmental protection efforts in computing, such as designing sustainable tech solutions and using computing resources responsibly to support long-term ecological well-being.
Equity (Fairness)	Ensuring fairness, justice, and equal opportunity in how computing affects people. This value calls for actively reducing biases and inequalities – for example, addressing the digital divide or algorithmic bias. It aligns with the principle of non-discrimination in tech, where ethical frameworks promote equity and fairness for all groups.
Growth	Embracing personal and collective development through computing. This value highlights using technology for positive progress – improving knowledge, skills, and quality of life. It also refers to societal development and economic growth driven by computing innovations, while continually learning and adapting.

Table 2.1: Codebook for SioC Values (*continued*)

Value	Definition
Integrity	Upholding high moral and ethical principles in computing – including honesty, rigor, and trustworthiness in one’s work. This value means doing the right thing even when no one is watching and maintaining credibility in computing practices.
Privacy & Confidentiality	Protecting sensitive information and respecting individuals’ privacy rights in the realm of computing. This combined value involves keeping personal or confidential data secure and not disclosing it inappropriately, thereby upholding trust. It reflects an understanding that computing innovations must balance data use with the right to privacy.
Respect	Treating others with dignity, fairness, and courtesy in all computing activities. It includes respecting people’s rights, diverse perspectives, and intellectual property, as well as not allowing technology to violate the respect due to individuals or communities.
Responsibility	Being accountable for the outcomes of computing and taking proactive ownership of one’s actions. It involves promoting benefits and avoiding harms in technology use, ensuring one’s work in computing does not negatively impact others.
Transparency	Openness and clarity about computing processes, decisions, and data. It involves disclosing methods, assumptions, and interests so that stakeholders can understand how technology works and decisions are made. Transparency builds trust by not keeping important information hidden.

With the qualitative coding complete, we constructed a concept network to visualize how these values interrelate in participants’ minds. In this network, each node represents one of the identified values. We added an edge (link) between two values if a single participant’s response was assigned both of those values – in other words, if those two values co-occurred in that person’s definition of social impacts. For example, if one participant talked about “the need for transparency and integrity in tech,” their response would contribute an edge between Transparency and Integrity. We repeated this for all 82 respondents, accumulating co-occurrences: the more frequently a pair of values was mentioned together by different people, the heavier (stronger) the edge connecting those two nodes. This approach follows the methodological inspiration of

Hoffman et al. (2014) who mapped individuals’ mental models as networks of co-mentioned concepts. In their study, “mental models are represented as networks where ... concepts are nodes, and links are established when a farmer mentions two concepts in their stated definition” of sustainability. Likewise, our network of values provides a systems view of the community’s collective “mental model” of social impacts of computing.

Using ‘igraph’ package (Csardi et al., 2024), we generated the network visualization using a force-directed layout (Kamada–Kawai algorithm) to spatially position values such that more strongly connected values appear closer together. We also computed basic network metrics (such as each

value's degree centrality) to identify which values were the most connected hubs in the network with the 'network' and/or 'sna' packages (Butts, 2010).

3. Network Analysis

The concept network (Figure 1) reveals a rich web of interrelated values that underpin how SioC workshop participants conceptualize the social impacts of computing. Several insights emerge from the structure of this network. First, certain value pairs co-occur very frequently, appearing together in many participants' responses. For instance, Responsibility and Transparency are tightly linked – a large number of respondents specifically mentioned the importance of being responsible in how we design/use technology and the need for transparency in tech's operations or outcomes. This is reflected by a prominently thick edge between Responsibility and Transparency in the network. Similarly, Responsibility–Integrity and Integrity–Transparency form strong connections, indicating that ethical integrity, transparency, and a sense of responsibility were often discussed in tandem. In fact, the densest cluster of connections centers on values related to ethical governance: Responsibility (accountability), Transparency (openness), and Integrity (honesty) form a triad of highly co-occurring values. These three values also had the highest centrality measures in the network (e.g. the greatest number of connections to other values). This suggests they serve as hubs in the collective value system – many participants linked one or more of these core ethical values to other facets of social impact. For example, a respondent might emphasize responsibility alongside inclusivity (“we have a responsibility to make tech accessible to all”), or transparency alongside environmental stewardship (“transparent reporting of tech's environmental impacts”). Thus, Responsibility, Transparency, and Integrity bridge to multiple themes, appearing as central connectors across the network.

In contrast, some values were mentioned less often and appear on the network periphery, with relatively few links. Notably, Equity and Environmental Stewardship emerged as two of the least-connected values in our data. Only a small subset of participants explicitly brought up equity (fairness in how computing's benefits and burdens are distributed) or environmental stewardship (the ecological responsibility of computing) in their definitions. These values have correspondingly thin or sparse connections in the network – for instance, Environmental Stewardship might link weakly to Responsibility (when someone noted responsibility for environmental impacts) but otherwise stands somewhat isolated. Inclusivity, which is related to equity, was mentioned by more participants and thus is a bit more central than Equity itself; still, inclusivity did not connect as widely as the top trio of ethical values. This relative scarcity of equity- and environment-related values in the discourse may reflect the particular makeup or priorities of the workshop attendees (more on this in the Discussion). It's important to note that the absence of strong links does not mean those values are unimportant overall, but rather that they were not top-of-mind for most participants when thinking of “social impacts of computing” – at least compared to values like transparency or responsibility.

With the cleaned data, we also examined the centrality metrics for each value node to quantify their roles in the network. The centrality analysis confirmed the visual impressions from Figure 1. Responsibility ranked highest on degree centrality (number of direct connections to other values), indicating it was linked to the widest variety of other values across participant responses. Transparency and Integrity were next, also scoring very high on centrality. These three values also tended

to have higher betweenness centrality (meaning they often lie on the shortest paths connecting other pairs of values in the network), reinforcing their role as bridging concepts. In practical terms, most participants – regardless of their specific interests (education, ethics, environment, etc.) – mentioned at least one of Responsibility, Transparency, or Integrity in their definition, making these values common meeting points in the overall conversation. By contrast, Equity and Environmental Stewardship had the lowest centrality measures – they were mentioned by fewer people and co-occurred with only a couple of other values. Values like Inclusivity, Adaptability, and Efficiency fell in a middle range: they were brought up by a moderate number of participants and linked to several other values, but not as pervasively as the big three. For instance, Inclusivity often co-occurred with Responsibility (e.g., responsibility to include diverse users) and with Integrity (perhaps in the sense of integrity to ethical principles of inclusion), but Inclusivity did not connect to as many distinct values as Transparency did. Overall, the network analysis paints a picture of which concepts the community consistently intertwines when thinking about computing’s social impact, and which concepts are more niche or specialized.

Table 3.1: Centrality metrics for each value

Value	Freq	IG deg	IG eig	IG bet	SNA deg	SNA eig	SNA bet
Responsibility	60	13	1.00	34.5	265	0.42	0.86
Transparency	49	13	0.97	10.0	247	0.41	0.86
Integrity	47	13	0.95	1.5	245	0.40	0.86
Collaboration	32	13	0.71	0.0	177	0.30	0.86
Curiosity	31	13	0.64	0.0	159	0.27	0.86
Privacy & Confidentiality	29	13	0.63	0.0	153	0.26	0.86
Caring	28	12	0.58	0.0	144	0.24	0.00
Respect	27	12	0.58	0.0	139	0.24	0.00
Adaptability	26	13	0.61	0.0	151	0.25	0.86
Growth	22	12	0.42	0.0	105	0.18	0.00
Citizenship	20	12	0.40	0.0	98	0.17	0.00
Efficiency	17	12	0.41	0.0	99	0.17	0.00
Environmental Stewardship	8	12	0.10	0.0	26	0.04	0.00
Equity (Fairness)	4	7	0.05	0.0	12	0.02	0.00

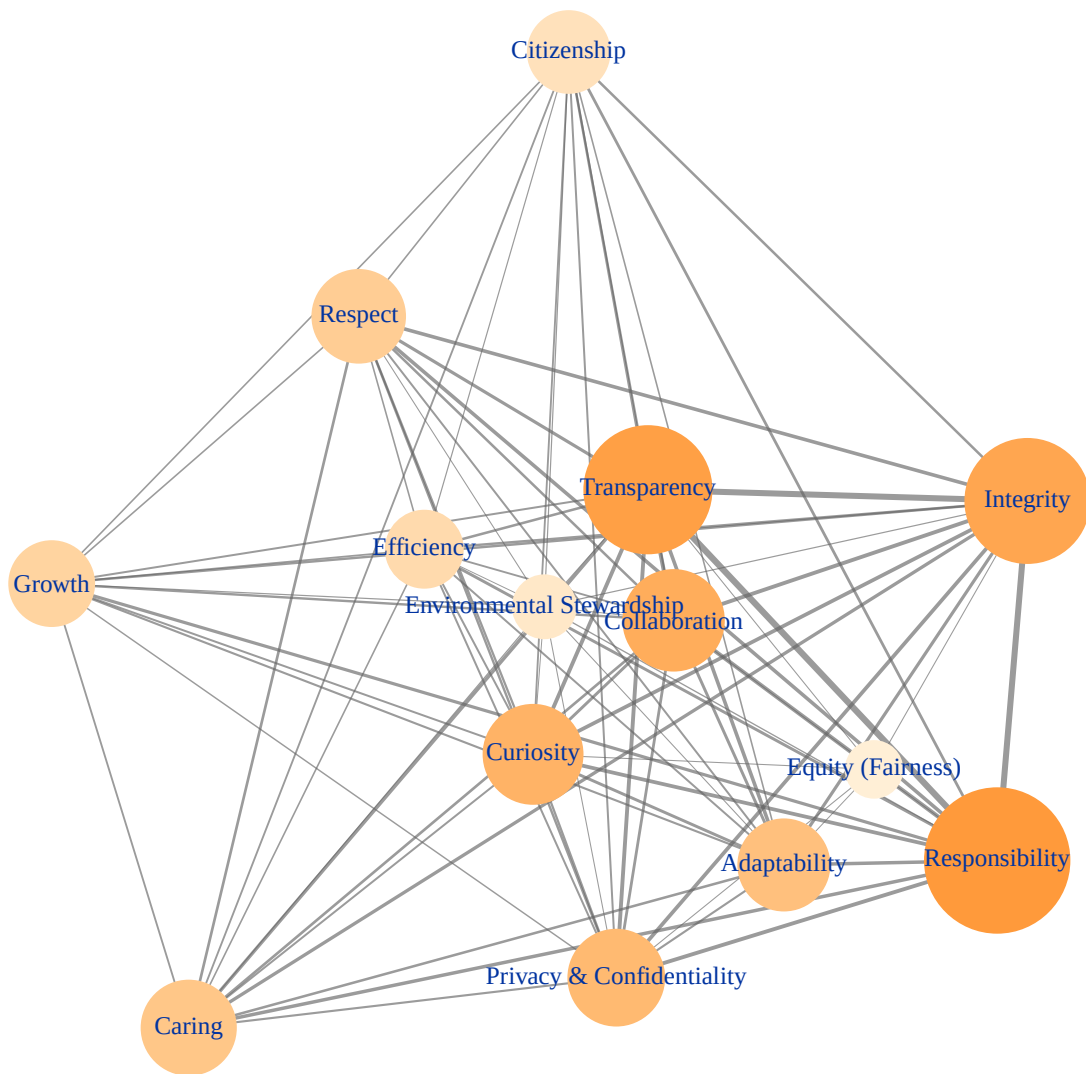


Figure 3.1: Concept network of values associated with “social impacts of computing,” based on co-occurrence in participants’ responses (Kamada-Kawai layout). Each node is a value (labeled), and edges connect values that were mentioned together by one or more respondents. Thicker edges indicate higher co-occurrence frequency, and larger nodes indicate values mentioned by more participants.

4. Discussion

The conceptual network of values offers a window into how this particular community – participants of the SioC 2025 workshop – understand the social impacts of computing. A key takeaway is that ethical accountability is at the heart of their collective understanding. The prominence of Responsibility, Transparency, and Integrity suggests that, to these stakeholders, computing’s social impact is fundamentally tied to questions of doing the right thing: ensuring technologies are developed and used responsibly, being transparent about their workings and consequences, and upholding integrity (honesty, trustworthiness) in computing practices. These values being most central means they were invoked across diverse contexts, whether one’s focus was on education, data, environment, or health. In essence, no matter which facet of tech’s impact someone was concerned with, they often anchored their thinking in ethical principles. This indicates a broad consensus that good governance and ethics in tech form the foundation of positive social impact. It aligns with the workshop’s emphasis on “emerging technologies that contribute to human flourishing” – such flourishing is perceived to require an ethical compass. Our findings here resonate with broader tech culture debates: for example, calls for AI ethics frequently highlight transparency and accountability as non-negotiable values (Gunder et al., 2025). What this network shows is that those calls are not just abstract ideals; they are actively shared by engaged community members when they reflect on computing’s role in society.

Beyond the ethical core, the network also highlights how participants weave together values from different domains, suggesting a holistic view of social impact. We see smaller clusters or linkages that correspond to more specific themes. For instance, one cluster connects values like Inclusivity, Equity, Respect, and Caring – values that all relate to how technology affects people in a social and cultural sense (ensuring everyone is included and treated with respect, and that technology is used to care for others’ well-being). This cluster aligns well with concerns in education and healthcare, two themes of the workshop. In education technology (XR and Ed Tech), inclusivity and respect are crucial for tools that serve diverse learners. In healthcare technology, caring and equity are key to ensuring tech improves patient outcomes without exacerbating disparities. Another grouping in the network brings together more innovation- and growth-oriented values like Curiosity, Adaptability, Efficiency, and Growth. These often appeared in responses that talked about computing driving social progress or personal development – for example, using computing to foster curiosity and learning, or the adaptability of society to technological change. Such ideas link to the XR/Ed Tech theme as well (curiosity and growth in learning), and even to organizational or economic impacts (efficiency and adaptability in workplaces). Meanwhile, the value Citizenship (which appeared in some responses) connected concepts of civic responsibility and digital citizenship, bridging the ethical domain with community engagement. The fact that our participants spontaneously mentioned Digital Citizenship or related ideas suggests an understanding that computing impacts not just individuals but the fabric of civic life – how we behave as members of society in a digital age.

The scarcity of certain values, like Equity and Environmental Stewardship, in the network is also revealing. It might initially seem surprising given that equity and environment are hot topics in broader tech and society discussions (e.g., calls for AI fairness and concerns about data centers’

carbon footprints). There are a few potential reasons why these came up less frequently in our study. One reason could be the phrasing of our question – by asking about “social impacts of computing,” participants may have intuitively focused on human social relationships and ethics, and less on environmental issues, which some might categorize separately as “environmental impacts” rather than social. Indeed, the workshop itself separated “Environmental Impacts” as a distinct theme, which might have led participants to think of environment as tangential when defining social impact. Another reason could be the composition of our respondents. It’s possible that relatively fewer participants in the survey were experts or advocates in environmental tech compared to, say, education or data ethics. If the majority of attendees were educators, ethicists, or healthcare professionals, they might naturally emphasize values aligned with their domains (like inclusion, transparency, privacy, etc.) and fewer had environment top-of-mind. Similarly, Equity might have been somewhat subsumed under Inclusivity in people’s minds – if both terms were on the checklist, some might choose one over the other. The lower mention of equity could also reflect that participants were thinking in general terms (like inclusion, fairness broadly) rather than the specific term “equity.” In any case, the relative underrepresentation of these values does not mean the community deems them unimportant; rather, it highlights an area where more dialogue might be needed. For example, one could argue that environmental stewardship is a social impact (since environmental effects ultimately impact communities). The workshop’s conversations might benefit from making that connection more explicit, ensuring that environmental values become a more integrated part of the social impact discourse in the future. Likewise, explicitly foregrounding equity – perhaps discussing digital equity, access gaps, or algorithmic bias – could enrich the community’s shared mental model so that equity joins integrity, transparency, and responsibility in the central value set.

Our findings underscore the importance of understanding culturally-situated values in any discussion of technology’s impact. Different communities or groups might have their own constellation of values that they prioritize. Richerson et al.’s work on cultural evolution provides a useful lens: they note that human groups exhibit massive cultural variation compared to other species, and these differences (in norms, values, etc.) can lead to different group outcomes. In the context of our study, the SioC workshop community can be seen as a cultural group – one that has coalesced around interest in the social implications of tech. The “selection” of Responsibility, Transparency, and Integrity as core values in this group’s dialogue suggests these are the traits that have proven salient and successful in the group’s discourse about positive tech futures. It’s a reminder that while frameworks for ethical tech often list many values (accountability, privacy, justice, sustainability, etc.), in practice people will gravitate to certain key values that resonate with their experiences and hopes. These core values emerge through collective conversation and consensus over time. In our case, through the workshop and related activities, participants implicitly negotiated what “social impacts of computing” means to them, and the network we mapped is essentially the output of that cultural discourse – a snapshot of shared understanding at this moment. As Gunder et al. (2025) suggest in the context of open education, when communities engage with technology, there is a “turn from tools alone to the practices and values that permeate and connect communities” (Gunder et al., 2025).

Our study makes those community values visible and explicit for SioC. It shows that shared values are not simply assumed or declared by experts – they are co-constructed by communities through dialogue, storytelling, and practice. This has an important implication: if we want ethical technology development to be effective, we must pay attention to and work with the values that specific user communities actually hold, rather than just imposing a one-size-fits-all value checklist.

Furthermore, the value network hints at which values might serve as entry points for policy and design in this community. Since Responsibility and Transparency are highly central, initiatives or technologies that exemplify these values might gain more traction or meet less resistance among stakeholders. For example, a proposal for a new AI system that includes clear transparency reports and accountability measures aligns with what this community cares about, and thus could be championed as addressing the social impacts they recognize. On the other hand, because Environmental Stewardship was not front-and-center, there may be a need for education or reframing to elevate its importance – for instance, drawing the connection that responsible computing includes responsibility to the planet. Likewise, the lack of emphasis on Equity might alert organizers to bring more voices or sessions that highlight equity issues (such as inviting experts in digital divide or algorithmic bias) in future workshops, to broaden the community’s perspective. In essence, mapping the values gives actionable insight into the current mindset and potential blind spots of the group.

5. Conclusion

This study demonstrated the value of using a network analysis of concepts to empirically map the values a community associates with the “social impacts of computing.” By surveying participants of the SioC 2025 workshop and analyzing how various values co-occur in their descriptions, we were able to visualize a collective mental model of what matters most to them in the realm of technology and society. The resulting concept network and centrality analysis illuminated a core set of ethical values (especially responsibility, transparency, and integrity) that act as the pillars of the community’s understanding, as well as secondary values that connect to specific contexts (like inclusivity in education or adaptability in innovation). This network-based approach moves beyond abstract ethical frameworks by revealing which values are actually salient to real stakeholders and how those values relate to each other in practice. Such insight is significant because it can guide the design of technology and policy to better align with the community’s values. When we know, for instance, that transparency and accountability are highly valued, designers and decision-makers can prioritize those features in computing systems or governance models, thereby enhancing acceptance and social benefit. Conversely, identifying values that are undervalued or overlooked (like environmental stewardship in this case) flags areas where additional community engagement or awareness-building might be needed.

In practical terms, our findings can inform the ongoing work of the Social Impacts of Computing group and similar initiatives. For educators, knowing the values students and faculty associate with computing can help shape curricula that resonate with those values (e.g. integrating responsibility and integrity themes into computing courses). For organizational leaders and policymakers, understanding this value landscape enables more culturally informed strategies – for example, crafting tech ethics guidelines that emphasize the community’s core values, or facilitating dialogues to bring less-mentioned values like equity to the forefront. The methodological approach we used is also broadly applicable. Future research can replicate this kind of values mapping in other communities or at a larger scale. By doing so, we could compare how different groups (tech industry professionals, end-users, policymakers, international communities, etc.) conceive of social impacts and where their value priorities converge or diverge. Such comparisons would be valuable for finding common ground and addressing conflicts in technology ethics debates.

There are several avenues to build on this work. First, increasing the diversity of participants in subsequent studies – including voices from various cultural, demographic, and professional backgrounds – would enrich the set of values and perhaps introduce new connections (for instance, a more globally diverse group might bring justice or community empowerment as key values). Second, future research could incorporate longitudinal data: measuring community values before and after an intervention (like the SioC workshop or an educational program) to see if discourse shifts values over time. Third, linking these self-reported values to actions or outcomes would deepen our understanding – for example, do participants who emphasize environmental stewardship in the survey also engage in more tech sustainability projects? Does a high centrality of transparency correspond to actual demands for open-source tools or explainability in AI systems within the

community? Investigating such links between values and behavior can help move from values identification to concrete impact. Finally, as we refine this research, developing better text analysis tools (with careful parsing, as our bracket mishap illustrated) and more interactive ways for communities to visualize and discuss their own values networks could empower groups to actively shape the narrative of what they believe “social impacts of computing” should entail.

In summary, mapping values conceptually has offered both clarity and direction: clarity in seeing the hidden structure of shared ideas that bind a community, and direction for driving ethical technology design that genuinely reflects community values. The SioC 2025 workshop set out to explore how emerging tech can support human well-being, and our analysis contributes to that goal by illuminating the value framework through which this community evaluates technology’s role in society. As technology continues to evolve, such grounded understanding of social impacts will be crucial for ensuring that innovation proceeds hand-in-hand with the values of the communities it is meant to serve.

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